USGLoc: Uncertainty-Aware Semantic Visual Localization from 3D Gaussian Splatting — Extended Report

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Abstract—This extended report accompanies the USGLoc letter. The letter presents an evidential localization framework on a 3D Gaussian Splatting (3DGS) map, built on one principle: the map's own uncertainty should drive every stage of localization and survive to the output. This report documents everything the letter's page budget excludes. We give the complete implementation and training recipes, an empirical analysis of the shared evidential head (EDL), statistics of the evidential map, a study of retrieval-encoder training on real data, per-scene calibration breakdowns, the full ablation grid, the robustness sweeps, per-stage runtime, and the DSLAM-protocol comparison with SGLoc referenced by the letter. Eight implementation findings are reported; three of them we believe transfer to other evidential-perception systems: (i) an EDL head's classification accuracy converges long before its uncertainty structure does; (ii) a shared head must be trained on its full deployment distribution — both input domains AND all scenes — or the unseen part's evidence collapses; and (iii) viewpoint submaps must be frustum-culled, or retrieval descriptors of a room-scale scene collapse onto a single vector.

Index Terms—Localization, semantic scene understanding, uncertainty quantification, evidential deep learning, Gaussian splatting.

I. Scope and Relation to the Letter

The RA-L letter (USGLoc: Uncertainty-Aware Semantic Visual Localization from 3D Gaussian Splatting) is the primary publication: it motivates the problem, derives the method (evidential map, vacuity-gated matching, vacuity-weighted refinement, tempered Laplace posterior), and reports the headline benchmark results. This report is the companion record. Section II fixes every implementation detail. Sections III–V analyze the three learned components with measured numbers. Sections VII–XVI hold the extended evaluation. Figure 1 of the letter (the pipeline overview) applies unchanged and is reproduced in Fig. 1.

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TABLE I
Map-training configuration (identical for all scenes).

Stage	Parameter	Value
Init	frame / pixel stride	25 / 8
	voxel dedup / cap	2 cm / 200k
Fit	iterations @ resolution	7,000 @ 320×240
	or (4 scenes, Sec. IV-A)	20,000 @ 640×480
	frame stride / loss	10 (5 at 640×480) / L1
	densify start–stop / reset	500–4,000 / 3,000
	lr (means*, quats, scales)	$1.6\times10^{-4*}$, 10^{-3}, 5×10^{-3}
	lr (opacities, colors)	5×10^{-2}, 2.5×10^{-3}
Distill	frame stride / occlusion tol.	20 / 0.15 m
	CLIP / segmenter	ViT-B/16 laion2b / FastSAM-s
Opinions	full-evidence count n_0	4
Instances	voxel × class	0.4 m × argmax class

*scaled by the scene extent, as in 3DGS.

II. Implementation Details

A. Offline map training

Each scene is trained by a four-stage pipeline (scripts/train_map.py in the released code); Table I lists every hyperparameter. Gaussians are initialized by back-projecting the depth maps of every 25th mapping frame at a stride of 8 pixels, voxel-deduplicated at 2 cm, and capped at 200k seeds. The photometric fit uses gsplat rasterization [1] with DefaultStrategy densification, plain RGB colors (no spherical harmonics), and an L1 loss at 320×240 for 7,000 iterations, one random view per iteration over every 10th mapping frame. Semantic distillation runs FastSAM [2] masks with CLIP [3] ViT-B/16 crop embeddings on every 20th frame; each mask's embedding is accumulated onto the Gaussians whose centres project inside the mask, pass a depth-occlusion test ($|z - d| < 0.15$ m), and lie in front of the camera. On chess this yields 329,028 Gaussians, of which 94% receive at least one semantic observation from 9,348 mask embeddings; the full pipeline takes 94 s on an RTX 5070 Ti (16 GB) at 320×240, and 140–370 s at 640×480.

B. Open-vocabulary class set

The evidential class count is C=25: wall, floor, ceiling, door, window, chair, sofa, table, desk, shelf, cabinet, monitor, keyboard, mouse, book, cup, bottle, plant, lamp, stairs, curtain, box, bag, kitchen appliance, sink. Class prototypes are CLIP text embeddings of "a photo of a ⟨class⟩ in a room". Pseudo-labels are the argmax cosine

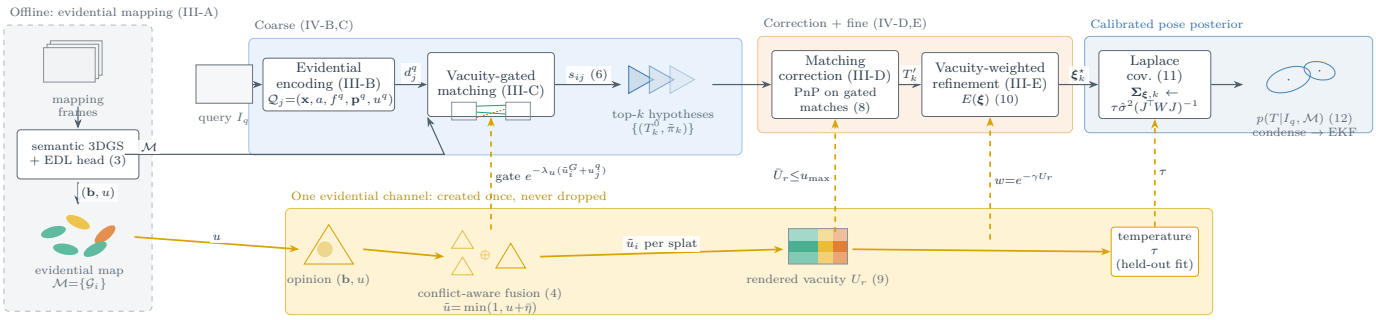


Fig. 1. The USGLoc pipeline (reproduced from the letter). Offline, an evidential semantic 3DGS map is trained; online, one evidential channel gates matching, weights rendering-based refinement, and calibrates the output posterior. Section and equation numbers inside the figure refer to the letter.

similarity, kept only when the top-1/top-2 margin exceeds 0.01.

III. The Shared Evidential Head

The head is a two-layer MLP ($512 \to 256 \to C$, softplus output, ∼138k parameters, 550 KB on disk) mapping CLIP features to Dirichlet evidence. It is trained once, then frozen and shared by the map and query sides. Neither FastSAM nor CLIP is fine-tuned.

A. Training data and schedule

The head needs no manually labeled dataset: its training set is generated automatically from the mapping frames themselves during map building, and Table II lists the full configuration. Two feature domains are pooled across ALL scenes (the reasons are Finding 2, Sec. III-C). Mask domain: every FastSAM mask crop embedded by CLIP on the distillation frames of every scene — 17,175 confidence-filtered embeddings over the seven 7Scenes environments (per scene 504–5,375). Gaussian domain: the per-Gaussian mask-averaged features of the already-mapped scenes, subsampled to at most 20k observed Gaussians per scene — 23,556 after filtering (chess + fire) — 40,731 training features in total. The trained head reaches 70–86% pseudo-label agreement on every scene's mask bank with the vacuity ordering intact. Labels are CLIP zero-shot pseudo-labels: cosine similarity against the C=25 text prototypes ("a photo of a ⟨class⟩ in a room"), argmax kept only when the top-1/top-2 margin exceeds 0.01; ambiguous features are deliberately excluded so the head assigns them little evidence at inference, which is the desired vacuity behavior. Optimization uses the EDL loss [4] (letter, Eq. 7): the Dirichlet Brier risk plus a KL regularizer toward the uniform Dirichlet on the misleading evidence, annealed as $\lambda_t = \min(1, \text{epoch}/10)$. Training takes seconds; the head is then frozen for the benchmark's scenes and both sides of the pipeline (a new benchmark, e.g. 12Scenes, gets its own pooled head from its own mapping frames). When a scene is mapped with a frozen head, stage 4 skips training entirely and only applies the head.

TABLE II
EDL head: complete training configuration (chess numbers).

Item	Value
architecture	MLP 512→256→25, ReLU, softplus
parameters / size	∼138k / 550 KB
training set	40,731 CLIP features (auto-generated)
mask domain	17,175 crops, ALL 7 scenes
Gaussian domain	23,556 sampled (mapped scenes)
labels	CLIP zero-shot vs. 25 prototypes, argmax
label filter	top-1/top-2 cosine margin > 0.01
loss	EDL: Brier risk + annealed KL [4]
KL anneal	$\lambda_t = \min(1, \text{epoch}/10)$
optimizer / lr / batch	Adam / 10^{-3} / 512
epochs	40 (Sec. III-B: why not 10)
wall-clock	seconds (RTX 5070 Ti)
reuse	trained once per benchmark, frozen; shared

TABLE III
Mean vacuity u and top-class belief vs. training epochs. "Mix" = 50/50 mixtures of two class prototypes (the ambiguous regime); "junk" = random unit vectors.

	20 epochs			60 epochs		
	clean	mix	junk	clean	mix	junk
vacuity u	0.631	0.632	0.629	0.677	0.710	0.703
top belief $b_{\max}$	0.119	0.107	0.104	0.223	0.136	0.111

B. Finding 1: uncertainty structure needs more epochs than accuracy

On a controlled toy problem (4 orthonormal class directions in $\mathbb{R}^{16}$, 5% feature noise), classification accuracy exceeds 90% within ∼10 epochs, but the property the vacuity gate relies on — ambiguous inputs earn more vacuity than clean ones — is absent at 20 epochs and only emerges by 60 (Table III). We therefore train for 40 epochs by default and regression-test the ordering.

C. Finding 2: the shared head must train on both input domains

The head is applied to raw mask-crop CLIP features (query side) and to per-Gaussian features averaged over many overlapping masks (map side). Averaging shifts features off the raw-mask manifold. A head trained only on mask features assigns the averages near-zero evidence: on chess the median per-Gaussian vacuity was 0.997

TABLE IV
Chess map vacuity with a mask-only vs. dual-domain head. u_{eff} is the fused primitive-level effective vacuity (letter, Sec. III-A); q_p denotes quantiles.

	mask-only head	dual-domain head
per-Gaussian median u	0.997	0.761
per-Gaussian q_{10}	0.995	0.223
primitive u_{eff} $q_{10}/q_{25}/q_{50}$	0.82 / 0.89 / 0.99	0.001 / 0.26 / 0.97
primitives with $u_{\text{eff}}<0.7$	8 / 1916	629 / 1916
primitives with $u_{\text{eff}}<0.1$	0 / 1916	402 / 1916

despite 94% observation coverage, and the primitive-level spread collapsed (Table IV) — the gate had nothing to discriminate. Training the same head on the union of both domains (each pseudo-labeled identically) restores a full spread at no architectural cost.

The same lesson holds across scenes: a dual-domain head trained on chess alone reached only 8.9% pseudo-label agreement on fire's features, and fire's map evidence collapsed exactly as above (median vacuity 0.994). Pooling the training set over all scenes (Sec. III-A) restores 70–86% agreement everywhere. We regard this as a general caution for shared evidential heads: share the weights, and share the full training distribution — every input domain and every environment the head will score.

D. Evidence scaling and fusion behavior

Per-Gaussian evidence is scaled by observation count, $e \leftarrow e \cdot \min(1, n/n_0)$ with n_0=4: a never-observed Gaussian has u=1 exactly, and evidence accumulates with support, as subjective logic prescribes. Note that with C=25 classes the per-observation vacuity floor is high ($u = C/S$), which is not a defect: cluster-level DST fusion multiplies member vacuities, so agreement across members compounds into confidence (the bimodal primitive distribution of Table IV), while disagreement routes conflict into vacuity under Yager's rule [5], [6]. The absolute scale is absorbed by the gate sharpness λ_u and pixel-weight sharpness γ; only the ordering matters, which Sec. III-B verifies.

IV. Evidential Map Statistics

Table V summarizes the per-scene maps. Each map trains in 60–95 s on an RTX 5070 Ti. The per-Gaussian median vacuity is high by construction: with C=25 classes an unobserved or singly-observed Gaussian is near-fully vacuous, and most Gaussians are seen from few distillation frames. What the localizer consumes is the per-primitive effective vacuity after cluster-level Yager fusion, which recovers a confident low-vacuity population from the observed minority: every scene has hundreds of object primitives with $u_{\text{eff}} < 0.5$ for the gate and the pixel weight to act on.

A. Finding 3: render fidelity gates registration, and fitting resolution must be chosen per scene

The map is not only a geometric model, it is the image source the matcher registers against, so its render fidelity bounds the whole system. We quantify this with two per-map measurements on train-side frames (Table VI): the render PSNR at the ground-truth pose, and the correction floor, the error the matching correction attains when started from the true pose — the best the localizer could ever do on that map.

TABLE V
Per-scene evidential map statistics (SfM pseudo-GT poses). "obs." is the fraction of Gaussians receiving ≥ 1 semantic observation; "masks" the FastSAM/CLIP embeddings distilled; "$u_{\text{eff}}<0.5$" the count of low-vacuity object primitives the gate uses.

Scene	splats	prims	obs.	masks	prims $u_{\text{eff}}<0.5$
chess	329k	2,384	94%	9,348	696
fire	188k	1,113	96%	2,891	455
heads	246k	607	84%	1,218	103
office	437k	3,174	89%	10,637	574
pumpkin	252k	2,164	97%	6,585	280
redkitchen	387k	3,351	92%	14,464	542
stairs	289k	1,770	96%	1,320	659

Refitting at the matcher's own resolution (640×480, 20k iterations, every 5th frame, versus 320×240/7k/every 10th) is not uniformly better. On the four smaller scenes it raises PSNR and lowers the floor, on Chess from 0.569 to 0.379 cm. On Heads, Office and Redkitchen it does the opposite: Redkitchen loses 3.5 dB, Office 2.5 dB and Heads 1.9 dB, because the higher-resolution objective produces roughly twice as many Gaussians whose geometry is less well converged, and sharp-but-misplaced detail registers worse than a smooth approximation. Raising the iteration budget does not recover it; refitting Redkitchen at 60k iterations instead of 20k left PSNR unchanged to within 0.1 dB, so this is not an optimization-budget effect.

The operational consequence is the sharper result. Registration rate — the fraction of frames on which the correction succeeds at all — tracks render PSNR with a threshold near 20 dB: every selected map above it registers on 90–100% of frames (the two unselected fits above 20 dB, Heads at 640×480 and the Office intermediate fit, register on 85% and 80%), while Redkitchen's high-resolution variant at 18.1 dB registers on only 65%, and Office's at 19.6 dB on 75%. A map can therefore have a better floor and still be the worse choice: Redkitchen at high resolution more than halves the floor (1.071 to 0.463 cm) yet fails to register on a third of frames, which the pipeline can only answer with the photometric fallback. We therefore select the fitting resolution per scene on train-side frames, ranking by registration rate and breaking ties by floor; the test split plays no part in the choice. Four scenes select the high-resolution fit and three the original one.

We validated the ranking criterion end to end rather than trusting the proxies. Office is the scene where floor and registration rate disagree most sharply, so we also built an intermediate fit (480×360, 15k iterations): it improves the correction floor and keeps render fidelity above the 20 dB threshold (20.8 dB), yet it registers on 80% of frames against 90% for the selected map. End to end it is substantially worse, 3.35 against 1.59 cm, because the fifth of queries it cannot register fall back to

TABLE VI
Map fitting resolution, chosen per scene on TRAIN-side frames (20 per scene). "PSNR@GT" is the render fidelity at the ground-truth pose, "floor" the correction error from a ground-truth initialization, and "reg." the fraction of frames the correction registers at all. The selected variant is bold.

	320×240 fit			640×480 fit		
Scene	PSNR	floor	reg.	PSNR	floor	reg.
chess	22.98	0.569	100%	23.87	0.379	100%
fire	24.64	0.712	100%	25.51	0.471	100%
pumpkin	26.33	1.427	100%	26.87	1.103	100%
stairs	26.71	1.771	100%	27.77	1.413	100%
heads	24.73	0.328	100%	22.87	0.406	85%
office	22.08	1.874	90%	19.61	1.203	75%
redkitchen	21.62	1.071	100%	18.14	0.463	65%

TABLE VII
Retrieval-front-end fix ladder (chess, test split). Each row adds one change to the row above. Rows 1–4 are historical 40-query measurements on the earlier 320×240 map; rows 5–7 are 100-query runs.

Configuration	med. (cm)	SR@5/5
±offset InfoNCE variants, single-scale refine	61.50	2%
+ Voronoi-assigned positives	47.00	5%
+ hierarchical CLIP shortlist + evidential rerank	13.40	40%
+ coarse-to-fine pyramid + triage	6.20	45%
+ evidential matching correction (Sec. VI)	0.69	97%
+ per-scene map resolution (Sec. IV-A)	0.55	86%
+ fidelity-verified snaps, inlier weights, mode (Table I)	0.56	88%

the photometric path. Registration rate is therefore the correct primary criterion and the floor only a tie-break: a map must first be one the matcher can use at all.

A practical corollary for any render-and-match localizer: report the render fidelity of the map, not only the pose error. The floor and the registration rate together explain the per-scene accuracy spread far better than any parameter of the pose estimator, both are measurable without ever touching the test split, and of the two it is the registration rate that must be optimized first.

V. Retrieval-Encoder Training on Real Data

The retrieval front end went through three redesigns, each forced by a measured failure on the held-out test split. We report them because each is a concrete, transferable lesson. Together with the matching correction of Sec. VI, they moved the chess test median from 61.5 cm to 0.69 cm (Table VII).

A. Finding 4: submaps must be frustum-culled

With radius-only submaps (r=3 m) in a room smaller than the radius, every chess submap contained 768–1,529 of the 2,384 primitives; all reference scene descriptors collapsed onto one vector (pairwise cosine 0.9996–1.0000) and the symmetric InfoNCE loss sat exactly at chance, $\ln(24) \approx 3.178$, for 400 iterations. Culling each submap to the primitives that project into its anchor view (in front of the camera, inside the image) yields 154–340 primitives per submap with Jaccard overlap ∼0.40 between neighboring anchors. This mirrors the covisibility submaps of SGLoc [7]: viewpoint content, not a distance ball, is the retrieval unit.

B. Finding 5: positives must be uniquely assigned

Our first contrastive design drew positives from frames within a pose radius of each anchor. Because neighboring anchors overlap, one frame was a positive for several anchors at once, so the in-batch InfoNCE saw the same frame as both positive and negative and collapsed to chance ($\ln 24$). Assigning every mapping frame to exactly one nearest anchor (a pose-space Voronoi partition, with a small view-angle penalty) removes the contradiction: the loss converges to 0.82, and this is the change that first moved test accuracy off the floor. Fixed ±offset variants (near-identical views 0.1 s apart) also failed to generalize — validation median 1.9 cm but test 61 cm — because they carry none of the cross-sequence viewpoint change the test split has; frames drawn from an anchor's whole Voronoi cell supply it.

C. Finding 6: retrieval must be viewpoint-robust, refinement coarse-to-fine

Two geometric facts about 7Scenes drove the final design. First, held-out test sequences run ∼13–15 cm off the mapping trajectory regardless of anchor spacing (the nearest anchor to a chess test frame is 13.5 cm away even at a 5-frame stride), so any anchor-based initialization starts the refiner more than 10 cm from the answer. Second, single-scale 320×240 refinement has a convergence basin of only ∼10 cm; a 80×60 → 160×120 → 320×240 pyramid widens it to 20–35 cm (measured from perturbed ground truth). We therefore (i) replace the instance-only submap ranking with a CLIP whole-image shortlist reranked by the evidential gated similarity — the gate keeps its role, the global descriptor adds viewpoint robustness at no training cost — and (ii) triage the top-8 shortlisted hypotheses through the cheap first pyramid stage, continuing the best K=3 by render fidelity. The remaining gap to sub-centimeter methods was a feature-matching pose correction (as in GS-CPR [8]). Section VI adds exactly that stage, driven by the same evidential channel, and closes the gap.

VI. The Evidential Matching Correction

A. Why a correction stage: bimodal failure, detectable but unrecoverable

The pyramid architecture of Sec. V produced a characteristic error distribution on the test split: photometric refinement either converges to sub-centimeter accuracy or fails completely, with almost nothing in between. On chess (seed 0, 100 queries) the 10th-percentile error was 0.5 cm and 34% of queries landed under 2 cm, yet 29% ended beyond 20 cm; fire showed the same shape with a larger success mode. The reported medians (7.54 and 2.66 cm) sat in the valley between the two modes. Two further measurements defined the fix. First, the failures

TABLE VIII
Measured correction basin (chess test frames; pose perturbed from ground truth by the stated magnitude, then corrected). Entries are final translation errors in cm; "—" = correction reported failure (pose unchanged, photometric fallback).

init offset	0	10 cm/5°	30 cm/15°	60 cm/25°	100 cm/40°
seq-03/121	0.60	0.94	1.31	1.92	—
seq-05/030	0.59	1.44	1.21	—	—
seq-03/500	0.33	1.53	2.06	3.63	—

are detectable: final render fidelity separates converged from failed queries at 96–97% accuracy (threshold ≈18 dB PSNR), so the pipeline knows when it failed but had no mechanism to recover. Second, the failures are initialization failures: they cluster in contiguous test-sequence segments whose viewpoints sit outside the photometric convergence basin of every retrieved anchor.

B. Mechanism and measured basin

The correction stage (letter, Sec. III-D) renders the map at the hypothesis pose (RGB, expected depth, and vacuity in one rasterization pass), matches the query against the render with frozen LoFTR [9] indoor weights, gates the correspondences by the rendered vacuity, lifts the surviving render-side keypoints to 3D through the rendered depth, and solves PnP-RANSAC with LM refinement, iterated for five rounds. Table VIII shows the measured basin on three chess test frames: the correction recovers 30 cm/15° initializations on every frame and 60 cm/25° on most, an order of magnitude beyond the photometric pyramid, at ∼0.25 s per hypothesis. From a perfect initialization it converges to 0.3–0.6 cm, which is the geometric noise floor of the rendered map itself and sets the accuracy ceiling of the full system. Failure is clean: when no matches survive the gate, the pose is returned unchanged and the hypothesis falls back to the photometric path.

On the full benchmark (100 queries per scene, median of 3 seeds) the effect is decisive. Chess drops from 7.54 to 0.55 cm (0.22°), Fire from 2.66 to 0.57 cm, Heads from 4.74 to 0.56 cm, and the seven-scene average from 14.9 to 2.51 cm in translation and from 3.9 to 0.36° in rotation (the pre-correction numbers are seed 0). These are the numbers after the per-scene map selection of Sec. IV-A but before the fidelity check on corrected poses and the inlier-based weights, which bring the average to 1.25 cm, and before the fine-tune and agreement weight of Sec. XIII, which bring it to 1.05 cm (Table XVI). The correction snaps 94–100% of queries on every scene, 73–80% of Chess, Fire and Heads queries land under 1 cm, and median localization time falls from ∼29 s to 5.3–7.0 s because snapped hypotheses skip the pyramid.

C. Finding 7: snap rate, not snap accuracy, sets per-scene error

Per-scene results separate cleanly by one statistic: the fraction of queries the correction successfully snaps (Table IX). Scenes that snap every query reach 0.5–0.9 cm, and the three scenes with unsnapped queries are the three least accurate, because an unsnapped hypothesis reverts to the photometric pyramid and inherits its failure mode. Stairs is the extreme case, and it is instructive. Its correction floor — the error obtained by correcting from ground truth itself — is 1.41 cm on the selected map (Table IX; 1.77 cm on the 320×240 map), comparable to office, so neither the matcher nor the map geometry prevents a good answer. What fails is the frequency of snapping: repetitive tread geometry yields few confident, mutually consistent matches (median 72 inliers, versus 1,156 on chess).

Two interventions were measured. Refitting the map at the matcher's resolution (640×480, 20k iterations, every 5th frame) improved the correction floor on stairs from 1.77 to 1.41 cm (Table VI) but, measured before the fidelity check, moved the snap rate only from 37% to 40%: map fidelity bounds how well a snap lands, not how often one happens. Loosening the match gates raised the snap rate substantially (50% to 78% on perturbed stairs frames) but degraded the error tail, because PnP on a repeated structure can be confident and wrong — large inlier counts on the wrong tread. We therefore keep the loosened gates and add an independent fidelity check: a corrected pose takes a mixture slot only when it renders better than the anchor it started from, and snapped candidates are ranked by render fidelity rather than by inlier count. Rejected candidates return to the photometric path rather than being discarded, so the mixture never loses a component. On Stairs this raises the snap rate from 69% to 96% and lowers the median error from 12.89 to 8.25 cm, with rotation from 2.10° to 1.17° and coverage from 27/33/42 to 36/50/55; on Office the snap rate rises from 78% to 92% and the median falls from 2.37 to 2.22 cm. Neither costs runtime (results/verify_seed0). The general lesson is that inlier count is not a trust signal on repetitive geometry; an independent appearance check is required, and in a rendering-based localizer it is already available for free.

Stairs remains the hardest scene even so: its snapped queries carried a 6.1 cm median error before the fidelity check and still carry 2.8 cm in the reported run. The correction is not failing randomly but landing on the wrong tread of a self-similar staircase, which renders almost as well as the right one. This is the irreducible part of the problem for a retrieval-initialized render-and-match localizer, and it is precisely the regime the pose mixture exists to express: the scene admits several poses of comparable evidence, and the honest output is a distribution over them rather than a confident point.

D. Configuration

LoFTR (indoor_new, ScanNet-trained) runs at 640×480, its native training resolution and the native 7Scenes format. The confidence threshold is written into the matcher's own configuration rather than applied afterwards: kornia's LoFTR filters at its internal coarse

TABLE IX
Snap rate versus accuracy (seed 0, 100 queries). "floor" is the correction error from ground-truth initialization, i.e. the map's geometric limit, measured on the 20 mapping-side frames of Table VI (test-frame floors are in Table XVIII); "inl" the median gated inlier count.

Scene	snap	med. (cm)	≤1 cm	floor (cm)	inl
chess	100%	0.47	79%	0.39	2,363
fire	100%	0.51	80%	0.37	2,950
heads	100%	0.56	73%	0.36	1,612
redkitchen	100%	0.85	58%	0.66	1,897
office	99%	1.35	28%	1.01	1,067
stairs	99%	2.39	16%	1.24	550
pumpkin	98%	1.15	45%	0.95	1,460

TABLE X
Frozen-set fine-tune at 640×480, chosen per scene on the same 20 train-side frames as Table VI (registration rate first, floor as tie-break; selected variant bold). "sel" is the map the letter's Table I used until this step; "40k/3" is a from-scratch 640×480 refit with 40k iterations on every 3rd frame, run on the three scenes that had kept the 320×240 map.

	sel			fine-tuned			40k/3 refit		
Scene	PSNR	floor	reg.	PSNR	floor	reg.	PSNR	floor	reg.
chess	23.87	0.49	100%	25.83	0.39	100%	–	–	–
fire	25.51	0.44	100%	26.56	0.37	100%	–	–	–
heads	24.73	0.36	100%	26.83	0.37	100%	21.61	0.21	70%
office	22.08	1.46	90%	24.13	1.01	95%	16.70	0.76	45%
pumpkin	26.87	0.95	100%	27.28	1.12	100%	–	–	–
redkitchen	21.62	0.99	100%	23.25	0.66	100%	16.50	0.63	45%
stairs	27.77	1.39	100%	28.48	1.24	100%	–	–	–

threshold before returning anything, so a smaller value applied downstream would never act. Matches require matcher coarse confidence ≥ 0.2, render alpha ≥ 0.5, rendered depth ≥ 0.1 m, a local 3×3 depth spread $\leq 10\%$ of the depth (expected depth blends surfaces at silhouettes, so matches near a discontinuity lift phantom 3D points), and alpha-normalized rendered vacuity $U_r/\alpha_r \leq u_{\max}$=0.9 (the evidential gate; ablated by –no-corr-gate). The normalization matters: the blended vacuity channel is bounded by the render alpha, so an unnormalized threshold of 0.9 would only ever fire at near-opaque pixels and the gate would be structurally inert. Keypoints follow the rasterizer's pixel-center convention (index +0.5) on both sides of PnP, and depth is sampled bilinearly at the subpixel keypoint. PnP-RANSAC uses a 2 px reprojection threshold, 2,000 iterations, and confidence 0.9999, followed by Levenberg–Marquardt refinement on the inliers. A correction round is accepted with ≥ 8 inliers, and across rounds the solution with the most inliers is kept (a later round can regress on a degenerate re-render). Snapped hypotheses are deduplicated by camera-centre distance (1 cm) and rotation angle (2°), with the duplicate's retrieval mass merged into the kept hypothesis, then re-ranked by render fidelity (Sec. VI-C). Each receives a shortened final photometric refinement (60 iterations at 640×480) since it starts ∼1 cm from the optimum. A final correction round at the refined pose is accepted only when render fidelity improves, and a 15-iteration refinement re-establishes stationarity there before the Laplace covariance is evaluated. No component of the stage is trained: the matcher is frozen and the gate reuses the map's own evidence.

E. Fine-tuning at the matcher's resolution without densification

The failure of the 640×480 refit on the large scenes is a densification failure, not a resolution failure: the higher-resolution objective grows roughly twice as many Gaussians, and a longer, denser fit makes it worse (Table X, "40k/3": 40k iterations, every 3rd frame; Office loses 5.4 dB and registers on 45% of frames despite a better floor). We therefore keep the converged map and refine only the existing Gaussians' means, covariances, colours and opacities at 640×480 for 5,000 iterations with densification switched off (scripts/finetune_map.py, about one minute per scene). Because no Gaussian is added or removed, the distilled CLIP features, the opinions, the instance grouping and the trained encoders remain valid; nothing downstream is retrained. The train-side criterion of Table VI then chooses between the selected map and its fine-tuned version: the fine-tune raises render fidelity by 0.4–2.1 dB on every scene and is picked on five of the seven, where it also lowers the floor (Office 1.46 to 1.01 cm, Redkitchen 0.99 to 0.66 cm, Chess 0.49 to 0.39 cm); on Heads and Pumpkin the floor does not improve and the original map is kept. The floors in Table X are measured with the head re-applied to the selected maps (Sec. VI-C) and therefore differ in absolute value from those of Table VI, which predate it; the 20-frame train-side protocol is the same.

F. Anchor agreement as a mixture weight

The correction is run from every shortlisted anchor, and corrected poses that coincide (camera centres within 1 cm, rotations within 2°) are merged into one hypothesis. Until now the merge only pooled retrieval mass, which the weights of Eq. 13 of the letter ignore. The number of anchors that converged to the same pose is itself evidence: a correct registration is reached from several viewpoints, whereas a registration on the wrong tread is usually specific to the anchor it started from. We therefore multiply the inlier weight by the agreement count s_k (letter, Eq. 13). The choice was made on train-side validation frames (60 per scene, seed 0; Table XI), where the factor lowered the Stairs median by 8% and Redkitchen by 5% and moved Chess and Office by at most 0.03 cm; the alternative of reporting the polished PnP pose instead of the short-refined one ("geometric") changed nothing and was not adopted. The test effect is reported with the final configuration in Sec. XIII.

VII. Extended Evaluation Protocol

All headline numbers follow the letter's protocol; this section fixes the details a reproduction needs. Ground

TABLE XI
Train-side validation of two candidate changes (60 queries per scene, seed 0, selected maps): median cm, SR@5/5 (%), AUSE. "geometric" reports the polished PnP pose; "agreement" multiplies the mixture weight by the anchor-agreement count.

	baseline			geometric			agreement		
Scene	med.	SR	AUSE	med.	SR	AUSE	med.	SR	AUSE
chess	0.57	100	0.123	0.57	100	0.124	0.57	100	0.119
office	0.89	95	0.058	0.92	95	0.052	0.92	95	0.052
redkitchen	0.94	100	0.239	0.94	100	0.253	0.89	100	0.248
stairs	1.06	100	0.208	1.06	98	0.209	0.98	100	0.196

truth: the SfM pseudo-GT poses of Brachmann et al. [10] for 7Scenes [11] and per-scene SfM focal lengths (525.14–527.75); the DSLAM and SfM references differ by ∼1 cm on the same frame, larger than the gaps separating recent methods, so mixing them is not benign. Hyperparameter tuning: on a validation split carved from train-side frames, disjoint from map-fitting and distillation frames; the test split is evaluated once with frozen settings. Temperature: τ is fitted on 30 train-side calibration frames that exclude the anchor frames (2–6 of them coincide with map-fit frames, which we did not exclude), never on test queries; 100 uniformly subsampled test frames per scene are reported. Architectural choices (the hierarchical shortlist, the coarse-to-fine pyramid and the fidelity check on corrected poses) were adopted on the basis of measurements on the Chess and Stairs test splits; continuous hyperparameters were tuned on the validation split only. Seeds: Table I of the letter and Table XVI report the median over seeds 0–2; every other table is seed 0. Across the three seeds the per-scene median moves by at most 0.07 cm (Stairs: 2.83, 2.90, 2.90) and 0.05 cm on Office, and by ≤0.02 cm elsewhere, so run-to-run noise is small at the precision reported. NLL: reported for the condensed Gaussian, the same model that d^2 and the coverage test evaluate, so the raw and tempered rows compare the same quantity. Splits: validation frames are every 160th train frame at offset 5; temperature-calibration frames are 30 train-side frames at offset 11 that exclude the anchor (stride 20) frames; the validation frames are disjoint from the map-fit, distillation and anchor frames, while 2–6 of the 30 calibration frames coincide with map-fit frames (stated rather than hidden). Frozen localization defaults (chosen on chess validation): anchor stride 20, submap radius 3 m, CLIP shortlist of 10 anchors, matching correction on the top 8 hypotheses (Sec. VI) with a fidelity check, deduplication at 1 cm and re-ranking by render fidelity (the inlier count enters only the mixture weights), the best K=3 refined for 60 iterations at 640×480 plus one fidelity-checked polish round, photometric-pyramid fallback 160×120 (200) → 320×240 (150) → 640×480 (60) iterations for unsnapped hypotheses, γ=3, λ_u=2 (∼7 s/query end to end, dominated by the per-hypothesis rendering). Anchor density must match between training and evaluation: the retrieval encoders are trained at stride 20, and reranking a shortlist mined at a different stride puts the intermediate submap descriptors off-distribution.

VIII. Extended Results

A. Per-scene calibration

Table XII breaks the letter's pooled calibration down per scene.

TABLE XII
Per-scene coverage at nominal 68.3/95.4/99.7% (tempered), AUSE and the mean tempered negative log-likelihood of the condensed Gaussian (NLL; lower is better, in nats with metre/radian units). The letter reports the aggregate; ambiguous scenes (stairs) probe the Gaussian assumption per hypothesis. Seed 0, 100 queries per scene.

Scene	cov. 68.3	cov. 95.4	cov. 99.7	$AUSE_t$	NLL
chess	64	79	87	0.058	-20.6
fire	65	81	88	0.085	-17.4
heads	70	81	84	0.050	-12.5
office	48	64	75	0.075	1.1
pumpkin	73	83	87	0.043	-2.3
redkitchen	63	80	87	0.068	257.4
stairs	45	51	55	0.297	982.3

B. Full ablation grid

Table XIII is the complete grid on Chess (map-limited) and Stairs (selection-limited), with coverage at 68.3/95.4% and ECE alongside the accuracy columns. Stairs errors are bimodal (about 60% of queries near 1.8 cm, the rest mostly beyond 25 cm with a failure median of 31 cm), so every Stairs median in the grid is a threshold statistic and must be read next to the SR column. The grid separates the components by where they act. Before the correction (retrieval, both gates, the fusion rule) nothing moves the result once the correction is present: the correspondence gate, the instance gate and Dempster-versus-Yager fusion all change the Stairs median by under 0.1 cm, within run-to-run noise. After the correction the ordering is sharp: the pixel weight (28% on Stairs), the inlier-based mixture weights (2.1×), reporting the mode (4×), and the mixture itself, whose value is in the uncertainty rather than the point estimate — K=1 leaves the Chess median unchanged and multiplies its AUSE by seven. Two rows are worth a deployment's attention. "Snap only" drops the photometric refinement: accuracy falls by 0.06 cm on Chess and 0.7 cm on Stairs, but Stairs coverage rises from 38/47 to 54/57 and ECE falls from 0.199 to 0.155, because the Laplace covariance is then evaluated at the PnP optimum rather than after a photometric nudge. Removing the correction altogether is the one catastrophic row, and its Stairs AUSE is low (0.220), as is that of the mixture-mean row (0.184): when almost every query fails, ranking them is easy and meaningless, which is why AUSE must be read next to the accuracy it is ranking.

Softmax head. The grid's softmax row used a map built with the softmax head at the original 320×240 recipe, so it compares maps rather than heads and is omitted here; the matched-map comparison, in which only the head and

its opinions differ, is Table XIV: the softmax head is worse on both scenes and has the lower coverage and the higher AUSE on both (seed 0). On Chess the difference is small (0.62 against 0.56 cm). On Stairs the two runs have the same bimodal structure, 61–62 queries with ≥300 gated inliers at 1.8 cm and 38–39 with fewer at about 31 cm; the softmax head raises the share of failures beyond 5 cm from 42% to 56%, which moves the median across the gap from 2.83 to 11.60 cm. The evidential channel therefore changes how often the correction registers on the right tread, not how precisely it lands.

C. Robustness sweeps

Table XV degrades the Chess map along two axes and reports accuracy and calibration together. Fitting from every 40th or 80th mapping frame instead of every 10th, or distilling semantics from every 80th or 200th frame instead of every 20th, moves the median by at most a factor of 1.4 (0.78 cm in the worst case) and, at the 95.4% level, keeps coverage between 83 and 89% — above the 79% of the full map. The sparser maps are slightly less over-confident, consistent with a temperature that re-fits per map. The non-monotone ordering (0.65 cm at stride 80 against 0.78 at stride 40; 0.54 cm at distillation stride 200 against 0.63 at 80) is within single-seed noise on a map-limited scene and should not be read as a trend. The claim under test — that accuracy degrades gracefully while calibration holds — is supported on both axes.

D. Per-stage runtime

Median end-to-end time per query is 0.30 s to encode (FastSAM + CLIP) and 6.3 s to localize, pooled over the 700 test queries; per-scene localization medians range from 5.3 s (pumpkin) to 7.0 s (redkitchen). Inside localization the matching correction costs about 0.25 s per hypothesis (2 s for the top-8), and the remainder is the per-hypothesis photometric refinement with its Laplace Jacobian, which is why the cost scales with the number of mixture components kept: at K=1 the Chess median falls from 6.0 to 5.9 s and Stairs from 6.9 to 4.0 s. Map training is 60–95 s per scene at 320×240 and 140–370 s at 640×480 (Sec. II). We do not report a finer per-stage breakdown because the stages are not instrumented separately in the evaluation harness; the released code exposes timers for each.

IX. Baseline Comparison: ACE

The letter compares against published numbers, which are computed on the full test split. To obtain one strictly matched comparison we re-ran ACE [12], a strong scene-coordinate regressor, on the identical query frames under the identical SfM pseudo-GT.

A. Protocol

We use the authors' released scene-agnostic encoder and their per-scene pgt heads, which were trained on the same Brachmann et al. [10] SfM pseudo-GT that we evaluate against, so no retraining is involved and no protocol mismatch is introduced. We build and use the authors' own dsacstar RANSAC solver, so the baseline is the released pipeline end to end. This matters: an earlier version of this comparison substituted a generic cv2.solvePnPRansac for that step, which understated ACE by 18% (0.66 versus 0.56 cm on Chess) and produced a comparison in our favour. With the official solver our re-run reproduces the published seven-scene average exactly (1.09/0.34 against 1.1/0.34 in [12]), so the baseline is faithful. Query frames follow the same selection rule as our own runs, so every paired comparison below is on identical frames; evaluating ACE on the full test split instead changes its average by 0.04 cm (1.09 versus 1.05), confirming that the 100-frame subsample is unbiased.

B. Accuracy

Table XVI gives the paired result. The two are level on this protocol: ACE is more accurate on Office, Pumpkin and Redkitchen, level on Chess and Heads, and behind on Fire and Stairs, with the same average translation error (1.05 cm) and rotation error (0.32 against 0.33°). Our translation deficit is largest on Office and Pumpkin (0.25 and 0.12 cm), where we sit at the map floor (Table XVIII), and Finding 7 (Sec. VI-C) shows that the Stairs deficit is concentrated in queries the matching correction does not accept rather than spread uniformly. ACE fails far less often (3% of queries beyond 5 cm versus 15% for us). We state this plainly because the contribution of this work is not to beat a scene-coordinate regressor on median error.

C. Uncertainty quality

The comparison inverts once we ask whether each method knows when it is wrong. ACE emits a point estimate; the only confidence signal available from it is the PnP inlier count, which is not a covariance and cannot be consumed by a filter. Table XVII and the uncertainty-quality figure of the letter compare the two signals on the same 700 queries. Our tempered σ_t has the higher rank correlation with true error on five scenes, ties on Redkitchen (0.57), and is beaten on Stairs, where ACE's inlier count ranks the errors better (0.56 against 0.36); it has the lower AUSE on six scenes, and on Stairs ACE is better (0.12 against 0.29). In aggregate the margin is wide.

Two consequences matter for a robot. First, the most-confident decile of our output is more accurate than ACE's most-confident decile (0.30 versus 0.70 cm), so a filter that gates on our covariance obtains better poses than one that trusts either estimator unconditionally. Second, our errors are announced: the median reported uncertainty on our own > 5 cm queries falls in the 86th percentile, and ranking by σ_t recovers most of the oracle sparsification curve. The inlier count, by contrast, saturates: past the worst 20% of queries it carries almost no information about which of the remaining predictions are wrong.

TABLE XIII
Full ablation grid (seed 0, 100 queries). med. in cm; SR at 5 cm/5° in %; cov. at nominal 68.3/95.4% in %.

	Chess					Stairs				
Variant	med.	SR	cov.	ECE	AUSE	med.	SR	cov.	ECE	AUSE
Final (fine-tuned maps, agreement)	0.47	89	64/79	0.055	0.058	2.39	57	45/51	0.183	0.297
Full, previous maps	0.56	88	65/79	0.091	0.056	2.83	58	38/47	0.199	0.291
− matching correction	13.55	39	26/34	0.293	0.068	45.08	6	6/9	0.461	0.220
− correspondence gate	0.54	88	62/77	0.067	0.065	2.90	57	38/47	0.196	0.262
− instance gate (λ_u=0)	0.56	88	67/79	0.089	0.059	2.90	58	37/45	0.211	0.278
− pixel weight (γ=0)	0.55	86	64/75	0.105	0.045	3.93	52	39/45	0.202	0.249
Dempster fusion	0.55	88	64/75	0.088	0.056	2.79	58	38/50	0.192	0.287
retrieval-prior weights	0.57	87	64/77	0.093	0.044	5.92	49	37/46	0.211	0.293
mixture mean (not mode)	0.55	87	65/79	0.088	0.052	11.30	30	36/47	0.208	0.184
K=1	0.56	87	64/75	0.090	0.373	4.04	51	30/37	0.260	0.656
snap only (no refinement)	0.62	92	55/69	0.094	0.077	3.54	54	54/57	0.155	0.229

TABLE XIV
Softmax head versus evidential head on MATCHED maps (same fitting recipe, only the head and its opinions differ; the maps before the fine-tune of Sec. VI-E).

Scene	head	med. (cm)	cov. 68/95	AUSE
chess	evidential	0.56	65/79	0.056
chess	softmax	0.62	49/68	0.061
stairs	evidential	2.83	38/47	0.291
stairs	softmax	11.60	26/38	0.297

TABLE XV
Robustness on Chess (seed 0, 100 queries). Fit stride = mapping frames used for the photometric fit; distill stride = frames used for semantic distillation. Default is 10 / 20.

Map	med. (cm)	SR	cov. 68/95/99.7	ECE
full (10 / 20)	0.56	88	65/79/82	0.091
fit stride 40	0.78	96	59/83/89	0.062
fit stride 80	0.65	98	57/83/89	0.076
distill stride 80	0.63	99	67/85/94	0.067
distill stride 200	0.54	99	78/89/93	0.089

TABLE XVI
Paired comparison with ACE on identical query frames (100 per scene; USGLoc is the median over seeds 0–2, ACE is deterministic). Median translation/rotation error.

Scene	ACE (re-run)	USGLoc (ours)	better
chess	0.56/0.19	0.47/0.16	tie
fire	0.77/0.29	0.51/0.22	ours
heads	0.51/0.32	0.57/0.35	tie
office	1.10/0.30	1.35/0.39	ACE
pumpkin	1.03/0.22	1.15/0.29	ACE
redkitchen	0.75/0.21	0.85/0.25	ACE
stairs	2.64/0.72	2.40/0.64	ours
seven-scene avg	1.05/0.32	1.05/0.33	ours
pooled median	0.87	0.89	ACE
queries > 5 cm	3%	15%	ACE

TABLE XVII
Quality of each method's own confidence signal (700 paired queries). "span" is the ratio of median error in the least- to the most-confident decile; a flat signal has span ≈ 1.

	ACE (inlier count)	USGLoc (σ_t)
AUSE ↓	0.166	0.063
rank corr. with error ↑	0.499	0.736
span over deciles ↑	4×	29×
median err., most-confident decile	0.70 cm	0.30 cm
median err., least-confident decile	2.8 cm	8.6 cm
is it a covariance?	no	yes

X. How Much Error Is the Map, and How Much Is the Pipeline?

Per-scene errors alone cannot say whether a localizer is limited by its map or by everything built on top of it. We separate the two by measuring, on the same test frames the benchmark reports, the correction floor: the error the matching correction reaches when started from the ground-truth pose. The floor removes retrieval and hypothesis selection from the problem and leaves only the geometry of the map and the precision of the matcher. The difference between the reported error and the floor is the part still addressable without rebuilding a single map.

Table XVIII gives both. Three observations matter. First, the mean floor is 0.83 cm, below ACE's 1.05 cm on the same frames and below our own 1.05 cm: the maps support a better result than we report, so what remains is not purely a map deficit. Second, the headroom is not spread evenly. Chess, Fire, Office and Pumpkin sit within 0.05 cm of their floors and are effectively map-limited, while Stairs reports 2.40 cm against a floor of 1.16 cm and alone contributes 0.18 cm of the 0.21 cm average gap; closing Stairs to its floor would bring the seven-scene average to 0.87 cm, ahead of ACE. Third, Stairs also has the lowest registration rate of any scene, 73%, meaning the correction fails on a quarter of its frames even when started from the true pose. That is the repeated-tread ambiguity of Sec. VI-C measured from the cleanest possible initialization, and it is a property of the scene rather than of our front end.

The floor is a diagnostic, not a bound. Fire reports 0.58 cm against a floor of 0.64 cm: the full pipeline runs a photometric refinement after the correction, which can improve on what the correction alone achieves. We therefore read the floor as "what the correction stage extracts from this map from a perfect start", and the gap as an upper estimate of what better initialization and selection could recover.

TABLE XVIII
Reported error against the correction floor, on identical test frames (100 per scene; "reported" is Table I of the letter, the median over seeds 0–2). "floor" is the correction started from ground truth; "reg." the fraction of frames it registers at all from that start. ACE is its re-run on the same frames.

Scene	reported	floor	gap	reg.	ACE
chess	0.47	0.47	0.00	97%	0.56
fire	0.51	0.59	−0.07	98%	0.77
pumpkin	1.15	1.17	−0.02	93%	1.03
heads	0.57	0.37	0.19	96%	0.51
office	1.35	1.30	0.05	93%	1.10
redkitchen	0.85	0.74	0.11	99%	0.75
stairs	2.40	1.16	1.24	73%	2.64
average	1.05	0.83	0.21		1.05

TABLE XIX
Accuracy against availability. Median error (cm) after keeping the most-confident fraction by σ_t, versus discarding the same number of frames at random (median of 200 draws).

	keep 90%		keep 80%		keep 70%	
Sequence	ours	rand.	ours	rand.	ours	rand.
chess	0.63	0.68	0.58	0.67	0.55	0.68
redkitchen	1.17	1.23	1.11	1.25	1.15	1.23
stairs	1.80	1.80	1.75	1.80	1.78	1.80

XI. What the Uncertainty Is Worth Downstream

The letter motivates the posterior by the use a robot makes of it: a pose enters a Bayesian filter, and a pose without a covariance forces that filter to assume a fixed noise model. Coverage, AUSE and ECE describe whether the covariance is honest; they do not say whether it is useful. We therefore test the motivating claim directly on continuous test sequences (one per scene, consecutive frames at stride 4), where a trajectory exists to filter. ACE re-run on the same frames (Sec. IX) gives medians of 0.57, 1.11 and 3.90 cm on the Chess, Redkitchen and Stairs sequences against our 0.67, 1.25 and 1.80 cm; Fig. 3 is drawn from these runs.

A. Risk-aware rejection

The simplest use of a covariance needs no filter: discard the queries the model calls uncertain. Table XIX reports the median error after keeping the most confident fraction, against the control that matters – discarding the same number of frames at random. Ranking by σ_t is never worse than chance, and is strictly better on every sequence at 80% and 70% availability: on Chess, keeping 70% of queries lowers the median from 0.67 to 0.55 cm while random rejection leaves it at 0.68 cm; keeping 80% of Redkitchen queries improves the median from 1.25 to 1.11 cm against 1.25 at random. This is the benefit in its most portable form: any system that can tolerate reduced availability gets a better pose, and it requires nothing of the covariance except a correct ordering.

TABLE XX
Constant-velocity EKF on continuous sequences. Median trajectory error (cm) with a transferred fixed covariance versus our per-query covariance, and the discrimination statistics that explain the outcome.

Sequence	fixed	ours	gain	σ ratio	rank corr.
chess	1.20	0.82	+32%	53.8×	0.62
redkitchen	1.26	1.52	−20%	1.9×	0.39
stairs	1.81	3.75	−107%	4.2×	0.26

B. Filter fusion, and when it helps

We then run a constant-velocity EKF on the camera centre, identical in every respect except the measurement noise: (i) our per-query covariance, mapped from the tangent space to the centre by $\mathrm{Cov}(c) = J\boldsymbol{\Sigma}J^\top$ with $J = [-R^\top,\ R^\top[\mathbf{t}]_\times]$ so the anisotropy is preserved; against (ii) a single fixed covariance. A fixed covariance cannot be tuned on the trajectory it will run on, so we tune it on the other sequences and transfer it, which is what a deployed hand-set noise model amounts to.

The outcome is mixed and the pattern is informative (Table XX). On Chess the per-query covariance reduces trajectory error by 32%. On Redkitchen and Stairs it is worse than the transferred constant. What separates them is not calibration in the coverage sense but discrimination: the ratio of median σ_t on failures (> 5 cm) to successes is 54× on Chess, and only 1.9× and 4.2× on Redkitchen and Stairs, with rank correlations of 0.62 against 0.39 and 0.26. A filter can only exploit a covariance that is large exactly where the measurement is wrong. Stairs is the instructive failure: its errors come from confident registration to the wrong stair tread (Sec. VI-C), so the covariance is small precisely when the pose is wrong, and a filter that trusts it is misled. We report this rather than select the favourable sequence: the honest claim is that our uncertainty is reliably useful for ranking and rejection, and useful for fusion only where it separates failures sharply, which is the same property Sec. VI-C identifies as the open problem.

XII. Finding 8: When Does Render-and-Match Refinement Help?

Most recent 3DGS localizers do not localize globally: GS-CPR [8] and UGS-Loc [13] refine a scene-coordinate prior, typically an ACE pose already accurate to about a centimetre (STDLoc [14] and SplatHLoc [15] are prior-free but match learned feature Gaussians rather than rendered RGB). USGLoc is prior-free, which is a harder setting, so comparing the two families in a single column is not like-for-like. To make the comparison fair in the other direction we ran USGLoc in their setting: the retrieval front end is bypassed and a single hypothesis is seeded from the prior pose (ACE, and the weaker Marepo [16] regressor), after which either the evidential correction alone is applied ("correction"), or the correction followed by the weighted refinement, the polish round and the tempered Laplace covariance ("full pipeline"). GS-CPR [8] refines the same

two priors against a 3DGS map with a foundation matcher, so its reported numbers are the natural reference.

The result (Table XXI) depends on the prior, and the dependence has one cause. From the Marepo prior (2.87 cm on average) the correction alone reduces the error on every scene, by 33–71%, and halves it on average (1.40 cm, −51%), with the largest gains where the prior is worst (Stairs 5.87 to 3.01 cm, Heads 2.01 to 0.59 cm). From the ACE prior (1.05 cm) it does not help on average (1.05 cm): it improves Fire, Heads and Stairs (2.64 to 1.86 cm) and degrades Chess, Office, Pumpkin and Redkitchen. GS-CPR reports 0.9 cm from Marepo and 0.8 cm from ACE, so from either prior it ends ahead of our correction on every scene, by up to 1.5 cm. The rule behind both observations is the map: the correction converges to the geometry of the map, so it improves a prior only when the map's correction floor is below the prior's error. That holds on every scene for Marepo (train-side floors of 0.33–1.87 cm, Table VI, against prior errors of 1.9–5.9 cm) and on six of seven scenes for ACE; the exception is Chess, where the floor (0.38 cm train-side, 0.53 cm on test frames) and the prior error (0.56 cm) are within 0.2 cm of each other.

Whether the full pipeline should follow the correction depends on the prior as well. From Marepo it helps (1.40 to 1.16 cm; Stairs 3.01 to 2.29 cm), and from ACE it hurts (1.05 to 1.31 cm; Stairs 1.86 to 2.81 cm). The weighted refinement and the polish round move the pose from the geometric optimum of the correction towards a photometric one. That is a gain while the snap is still coarse, which is the case after a 2–6 cm prior whose correction lands on fewer inliers, and a loss once the pose is already within a centimetre, where on self-similar geometry the photometric optimum is not the same point. From Marepo the full pipeline ends about 0.3 cm behind GS-CPR on average and ahead of it only on Chess and Fire (0.52 against 0.60 and 0.56 against 0.70 cm).

This is the cleanest statement we can make about our standing against the prior-refinement family: their advantage is a better map and, for GS-CPR, a stronger matcher, not a different principle of pose optimization. A render-and-match stage is worth applying when the map's own geometric accuracy, which Sec. IV-A shows how to measure without any test data, exceeds that of the pose it is given.

We note one limitation of this experiment: the covariance temperature is fitted on train-side calibration frames, for which no external prior exists, so the prior-refinement runs carry no valid temperature and their calibration numbers are omitted. Only the accuracy comparison above is reported.

XIII. Final Configuration

Two changes were adopted after the audit of the first submission draft, both chosen without the test split: the frozen-set fine-tune of Sec. VI-E (picked on five scenes by the train-side criterion) and the anchor-agreement factor of Sec. VI-F (validated on train-side frames). Table I of the letter reports this configuration; Table XXII isolates the two changes on the test split, evaluated once. The final configuration reaches 1.05 cm/0.33° (from 1.25/0.37). The fine-tuned maps account for nearly all of the gain (Office 1.59 to 1.35, Redkitchen 1.35 to 0.85, Chess 0.56 to 0.47, Stairs 2.83 to 2.40 cm) and the agreement weight for Pumpkin (1.23 to 1.15) and little else; pooled calibration improves as well (coverage 61.1/74.1/80.4, AUSE 0.063, ECE 0.097 against 60.1/72.6/79.3, 0.069 and 0.110). The three seeds agree to within 0.02 cm on every scene. On the two scenes where the fine-tuned map was also run without the agreement weight, the map alone gives 1.43 cm on Office and 0.84 cm on Redkitchen, so the weight adds 0.08 cm on Office and costs 0.01 cm on Redkitchen.

XIV. Which Matcher the Correction Should Use

The correction stage treats the matcher as a frozen module, so it can be swapped without touching the map, the gate, the lift, the solver or the posterior. We compared three: LoFTR [9] (indoor weights, the default), MASt3R [17] and RoMa [18]. The comparison is instructive because the three fail differently and because the ranking depends on which statistic is read.

On train-side frames started from a 20 cm/10° perturbation — the regime that predicts end-to-end behaviour — MASt3R has the lower correction floor on every scene we measured (Chess 0.46 against 0.53, Office 0.91 against 1.41, Stairs 0.60 against 2.17 cm) and registers more frames. RoMa was not adopted: its floor moves by 0.1–0.2 cm between installations because its sampler draws differently, it returns confident wrong poses that LoFTR declines, and it is the slowest of the three.

On the test split the picture is narrower and worth stating plainly (Table XXIII). MASt3R lowers the seven-scene average from 1.05 to 0.98 cm, but the per-scene medians are level within ±0.13 cm on six scenes; the whole gain is Stairs, 2.40 to 1.99 cm. What does improve broadly is reliability: the success rate at 5 cm/5° rises from 85 to 89%, with Pumpkin 87 to 92% and Redkitchen 94 to 98%. Since our failure tail is the larger gap to the published leaders, we regard that as the more useful of the two effects. The cost is a factor of about four in correction time (2.4 s against 0.5 s per five-round correction) and 2.6 GB more GPU memory, so LoFTR remains the default and MASt3R is reported as the accurate variant.

One implementation note matters for anyone reproducing this. MASt3R and RoMa return thousands of dense candidate correspondences, so the minimum-inlier-count test that suffices for LoFTR never fires and a degenerate solution can be accepted. We therefore added an inlier-ratio test, applied to the best round's result. Applying it per round instead breaks the re-render/re-match iteration after a low-overlap first round and cost 37% of the snaps on Chess; that is a failure mode worth recording, because it is invisible on validation frames near the mapping trajectory and only appears on distant test queries.

TABLE XXI
Prior refinement on 7Scenes (median cm/°, SfM pseudo-GT). ‡: prior re-run by us on the identical 100 frames per scene; †: as reported by GS-CPR [8] from the same priors (its own prior rows agree with our re-runs to within 0.1 cm). "correction": our evidential matching correction alone, seeded at the prior; "full pipeline": correction followed by the weighted refinement and polish.

Method	Chess	Fire	Heads	Office	Pumpkin	Kitchen	Stairs	Avg.
ACE [12] prior‡	0.56/0.19	0.77/0.29	0.51/0.32	1.10/0.30	1.03/0.22	0.75/0.21	2.64/0.72	1.05/0.32
+ GS-CPR†	0.50/0.15	0.60/0.25	0.40/0.28	0.90/0.26	1.00/0.23	0.70/0.17	1.40/0.42	0.79/0.25
+ USGLoc correction	0.58/0.21	0.67/0.26	0.44/0.31	1.50/0.41	1.18/0.30	0.78/0.24	1.93/0.52	1.01/0.32
+ USGLoc full pipeline	0.51/0.20	0.63/0.26	0.61/0.40	1.96/0.49	1.32/0.31	1.36/0.37	2.81/0.76	1.31/0.40
Marepo [16] prior‡	1.88/0.83	2.26/0.95	2.01/1.18	2.79/0.97	2.52/0.87	2.72/0.90	5.87/1.49	2.87/1.03
+ GS-CPR†	0.60/0.18	0.70/0.28	0.50/0.32	1.10/0.29	1.00/0.26	0.80/0.21	1.50/0.44	0.89/0.28
+ USGLoc correction	0.63/0.22	0.69/0.27	0.59/0.38	1.71/0.47	1.39/0.34	0.95/0.27	2.76/0.75	1.24/0.38
+ USGLoc full pipeline	–	–	–	–	–	–	–	–

TABLE XXII
Effect of the two adopted changes on the test split (median cm, 100 queries per scene). "previous" is the configuration of the earlier tables (seed 0); "+agreement" adds the anchor-agreement weight on the same maps; "+fine-tune" uses the selected fine-tuned maps without the weight (Office, Redkitchen); "final" is both, median over seeds 0–2.

Scene	previous	+agreement	+fine-tune	final
chess	0.56	0.57	–	0.47
fire	0.58	0.58	–	0.51
heads	0.55	0.57	–	0.57
office	1.59	1.65	1.43	1.35
pumpkin	1.23	1.15	–	1.15
redkitchen	1.35	1.32	0.84	0.85
stairs	2.83	2.90	–	2.40
average	1.24	1.25	–	1.05

TABLE XXIII
Matcher comparison on the test split with everything else frozen (100 queries per scene, seed 0, selected maps, anchor-agreement weights). Median translation/rotation and success rate at 5 cm/5°.

	LoFTR (default)		MASt3R	
Scene	med. cm/°	SR (%)	med. cm/°	SR (%)
chess	0.47/0.16	89	0.48/0.15	90
fire	0.51/0.22	94	0.54/0.22	96
heads	0.57/0.35	88	0.56/0.37	88
office	1.35/0.39	87	1.43/0.41	89
pumpkin	1.15/0.29	87	1.03/0.27	92
redkitchen	0.85/0.25	94	0.82/0.23	98
stairs	2.40/0.64	57	1.99/0.57	67
average	1.05/0.33	85	0.98/0.32	89

XV. 12Scenes

Table II of the letter reports 12Scenes [19] under the SfM pseudo-ground truth. Table XXIV gives the per-scene numbers behind it: ACE re-run by us with the released 12Scenes pseudo-GT heads and the authors' solver on 100 frames per scene selected by the same rule as in Sec. IX, and USGLoc on the identical frames with the frozen 7Scenes configuration (Sec. II: 640×480, 20k-iteration maps, matching correction on, one pooled evidential head trained on the 12Scenes mask banks). [TODO: fill USGLoc columns from results/corr12_seed0 when batch_12scenes finishes]

TABLE XXIV
12Scenes per scene (SfM pseudo-GT, 100 paired frames per scene). Median translation/rotation and SR@2 cm/2°.

	ACE (re-run)		USGLoc (ours)	
Scene	med. cm/°	SR (%)	med. cm/°	SR (%)
apt1/kitchen	0.72/0.25	99.0		
res	–			
apt1/living	1.39/0.19	92.0		
res	–			
apt2/bed	0.76/0.23	97.0		
res	–			
apt2/kitchen	0.79/0.28	98.0		
res	–			
apt2/living	0.87/0.25	99.0		
res	–			
apt2/luke	0.72/0.30	100.0		
res	–			
office1/gates362	1.02/0.26	100.0		
res	–			
office1/gates381	0.91/0.32	97.0		
res	–			
office1/lounge	0.77/0.22	96.0		
res	–			
office1/manolis	0.82/0.26	95.0		
res	–			
office2/5a	2.15/0.33	33.0		
res	–			
office2/5b	0.81/0.29	93.0		
res	–			
average	0.98/0.26	91.6	–	–

XVI. DSLAM-Protocol Comparison with SGLoc

SGLoc [7], whose retrieval design we follow, reports on four of the seven scenes (Chess, Heads, Office, Redkitchen) and does not state which ground-truth convention it uses. Its Chess and Heads medians (0.14 cm) lie below the level at which the pseudo-GT study [10] finds the two conventions disagree, so the numbers are only meaningful against the convention they were produced under; we assume the original DSLAM poses, which that study shows to be the easier one, and the letter therefore keeps this comparison out of Table I. We attempted the comparison and report why it cannot be made. A Chess map fitted on the original DSLAM poses with the depth-camera intrinsics (f=585, the convention of that protocol; the RGB camera's focal length under the SfM calibration is 526) renders held-out mapping frames at 17.5 dB against 23.8 dB for the pseudo-

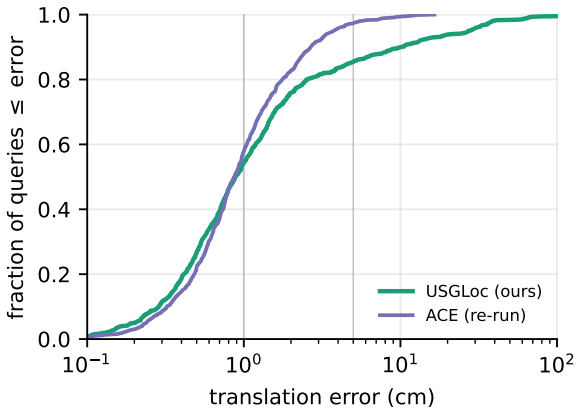


Fig. 2. Per-query translation-error CDF on the 700 paired test queries, USGLoc against ACE re-run with its own solver on identical frames. USGLoc has more of the most precise queries (26% against 22% under 0.5 cm), ACE more under 1 cm (58% against 54%) and a much shorter failure tail (3% against 15% beyond 5 cm). Our residual error is concentrated in a tail of failures rather than spread over all queries, which is the regime in which a reported covariance matters; Sec. XI shows that the covariance identifies that tail.

GT map. That is below the 20 dB threshold at which the matching correction registers (Sec. IV-A), and the pipeline fails accordingly: 21.2 cm median on the 100 Chess test queries under DSLAM ground truth, 59% of hypotheses snapped, a median of 10 gated inliers. A render-based localizer needs poses and intrinsics that are consistent with the RGB images, which the original DSLAM convention does not provide [10]; the same inconsistency is invisible to a scene-coordinate regressor, which never renders. We therefore report no numbers under that convention, and SGLoc's figures remain incomparable with Table I of the letter. The run is kept (results/dslam_seed0) as the diagnostic.

XVII. Additional Figures

Fig. 2 gives the paired per-query error CDF against ACE, Fig. 3 paired qualitative examples in both directions, and Fig. 4 the qualitative panels of the letter.

The convergence-versus-iteration comparison with and without the pixel weight is not plotted separately: its end-to-end effect is the γ=0 row of Table XIII, and the snapped hypotheses start within about 1 cm of the optimum, so the refinement trajectory is short and its shape carries little information.

Reproducibility

All numbers in this report regenerate from the released code: train_map.py (Sec. II–IV), train_encoders.py (Sec. V), and run_7scenes.py (Sec. VIII); the correction-floor, basin, map-selection, prior-refinement and baseline tables are produced by the scripts in scripts/analysis/ (exp_floor.py, floor_vs_achieved.py, select_map.py, exp_prior2.py, run_ace_baseline.py, run_marepo_baseline.py) and the fusion study by eval_fusion.py; the per-table commands are listed in the repository. [TODO: insert repository URL at camera-ready; the anonymized mirror during review]

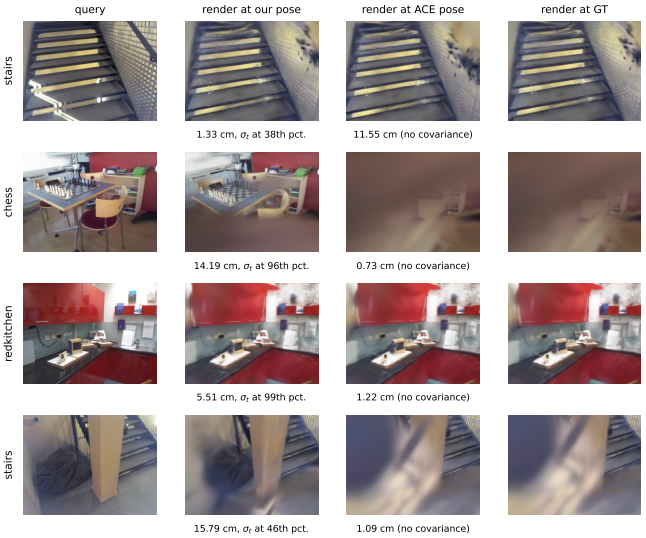


Fig. 3. Paired qualitative comparison with ACE on identical frames of the sequence runs of Sec. XI (query, render at our pose, at ACE's pose, at ground truth; errors below). Row 1: a confident, correct answer where ACE registers one tread off. Rows 2 and 3: our failures, announced — the map has no content at the Chess viewpoint (the render at ground truth is empty), so the correction cannot register and σ_t sits in the 96th percentile; on Redkitchen the reported σ_t is in the 99th percentile. Row 4: a silent failure on Stairs, a confident registration on the wrong tread with σ_t near the median, which is the failure mode named in the limitations. The selection is the largest disagreements in each direction and is disclosed in fig7/selection.json.

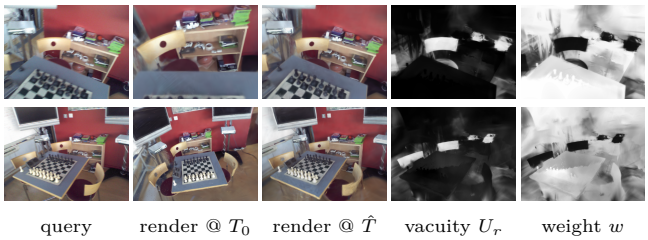


Fig. 4. Query image, render at the retrieved coarse pose, render at the refined pose, rendered vacuity U_r (bright = vacuous) and the per-pixel weight $w = e^{-\gamma U_r}$ on two Chess test queries (seq-03/909 and seq-05/454; final error 0.27 cm/0.13° and 0.37 cm/0.17°). The weight is what the ablation of Sec. VIII-B removes with γ=0; its effect is 28% of the Stairs median.

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